

	Case Name: Staff Authorization System	Sector	IT
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions
Submitted by	Matilde Casado and Margarida Antunes		
Date	2015	Project Control	Automatic tracking
File Name	C2015-11 Staff Authorization System		Tracking based on user input

1. Project description

Project authenticity

A Staff Authorization System project involving requirements gathering, analysis, development, validation, and implementation for secure and efficient access control.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	13
Planned Duration (PD)	55 days*
Budget At Completion (BAC)	14.400 €
Renewable Resources	5
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	25%
Activity Distribution (AD)	66%
Length of Arcs (LA)	0%
Topological Float (TF)	52%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	17.3	23.5	-1.4
CRI-rho	43.7	15.5	-0.1
CRI-tau	63.6	42.3	0.4

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	10.0	23.5	-2.3
CRI-rho	10.0	23.5	-2.3
CRI-tau	10.0	23.5	-2.3

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	29.5	41.6	-0.9
SI	29.1	37.5	-0.9
SSI	12.5	27.7	-2.3
CRI-r	12.6	21.4	-2.2
CRI-rho	38.7	20.6	0.5
CRI-tau	61.8	44.1	0.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	339.7	-339.7	1	3.7	0.7
PV - SPI	253.5	-253.5	CPI	3.8	1.8
PV - SCI	244.0	-244.0	SPI	14.2	14.2
ED - 1	502.3	-502.3	SPI(t)	14.8	14.8
ED - SPI	252.5	-253.5	SCI	14.4	14.4
ED - SCI	245.7	-245.7	SCI(t)	15.0	15.0
ES - 1	352.5	-352.5	0.8 CPI + 0.2 SPI	6.6	6.4
ES - SPI(t)	104.9	-104.9	0.8 CPI + 0.2 SPI(t)	6.9	6.8
ES - SCI(t)	102.2	-102.1			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	4240.0	-1485.0	-29.7	1.6	0.9	0.8	0.8
std dev	1897.9	2538.4	50.5	0.1	0.2	0.4	0.2
final	5295.0	0	0	1.6	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	55 days
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Real Duration	56 days
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Late	1.8%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	73.0	1.0	10.1	-10.1
PV - SPI	74.0	1.0	10.7	-10.7
PV - SCI	74.7	0.6	11.1	-11.1
ED - 1	75.0	0.0	11.3	-11.3
ED - SPI	75.3	0.6	11.5	-11.5
ED - SCI	76.0	1.0	11.9	-11.9
ES - 1	77.0	1.0	12.5	-12.5
ES - SPI(t)	78.0	1.0	13.1	-13.1
ES - SCI(t)	79.0	1.0	13.7	-13.7

2.3.3.3. Cost forecasting

BAC	14.400 €
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Real Cost	9.105 €
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Under Budget	-36.8%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	10160.0	1897.9	3.9	-3.9
CPI	9059.6	454.5	0.2	0.2
SPI	12216.6	5456.9	11.4	-11.4
SPI(t)	16314.7	12554.4	26.4	-26.4
SCI	10414.9	2730.5	4.8	-4.8
SCI(t)	13115.6	7400.1	14.7	-14.7
0.8 CPI + 0.2 SPI	9339.7	840.6	0.9	-0.9
0.8 CPI + 0.2 SPI(t)	9426.8	986.3	1.2	-1.2